BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 116 · Parameters, Return Values and Scope

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How is a cell selected in a two-dimensional array?

Q2. What should a useful code comment explain?
A. The programmer's password
B. An unrelated topic
C. A misleading instruction
D. Why the code is needed

Q3. “The CPU's main purpose is permanent file storage.” Correct this. Explain why.

Q4. A folder contains 20 images of 200 KB each. Using decimal units, give its total size in KB and MB.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Parameter — A value passed to a subprogram when it is called.

Return value — A function sends a result back to the calling code.

Local scope — A local variable is accessible within its subprogram.

Global scope — A global variable is defined outside subprograms and can be accessed more widely, increasing risk of unintended changes.

New learning

Parameter: A value passed to a subprogram when it is called.

Return value: A function sends a result back to the calling code.

Local scope: A local variable is accessible within its subprogram.

Global scope: A global variable is defined outside subprograms and can be accessed more widely, increasing risk of unintended changes.

Worked example

Trace a function double(x) that returns x*2 when called with 7.

1. The argument 7 is passed into parameter x.

2. Inside the function x*2 gives 14.

3. The function returns 14 to its caller.

4. If printed, the output is 14.

We Do · Practise with guidance

Guided 1

Trace a function addTax(price) which returns price+2 for input 9.

Worked method: Return value 11.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the line using the global number of bits per byte.

```
global bitsPerByte = 8
function toBits(bytes)
    totalBits = ...
    return totalBits
endfunction
print(toBits(7))
```

write

Question 1

Differentiate a function and a procedure. [2 marks]

Question 2

A function returns x+5. What is returned for x=8? [1 mark]

Question 3

Explain one benefit of local variables. [2 marks]

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

A local variable is necessarily accessible from every subprogram.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Use two indexes: one for the row and one for the column.

Q2. D — Why the code is needed

Q3. It executes instructions to process data.

Q4. (a) 4,000 KB = 4 MB.

I Do / We Do pages · Guided practice

Q1. Return value 11.

You Do pages · Code First

```
global bitsPerByte = 8
function toBits(bytes)
    totalBits = bytes * bitsPerByte
    return totalBits
endfunction
print(toBits(7))
```

Expected result: 56

You Do pages · Independent questions

Q1. A function returns a value; a procedure performs an action and does not need to return a value.

Q2. 13.

Q3. They limit unintended interference between subprograms and make code easier to reason about.

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: